

Arithmetic Statistics from –163 to the Present

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Colloque des Sciences Mathématiques du Québec
`thornef.github.io/montreal-colloquium.pdf`



An outlandish question

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- ▶ ...

$41^2 + 41 + 41$ is **definitely not** prime.

$41^2 + 41 + 41$ is **definitely not** prime.

But: $n^2 + n + 41$ **is** for $0 \leq n \leq 39$.

*Six sensational discoveries that somehow
or another have escaped public attention*

As the steady rise of public interest in occultism and pseudoscience over the past 10 years in the U.S. been something apart from public understanding of scientific knowledge? The two have interacted more strongly than most people realize. Important advances in science have been crowded out of newspapers, magazines, radio and television to make room for reports on poltergeists, demon possession, psychic healing, prehistoric visits to the earth by astronauts from other worlds, the vanishing of ships and planes in the Bermuda Triangle, the emotional life of plants, the primal scream and so on and on nauseum.

Coxeter's insight has now been vindicated. Last November, William McGregor, a graph theorist of Wappingers Falls, N.Y., constructed a map of 110 regions that cannot be colored with fewer than five colors [see upper illustration on page 128]. McGregor's technical report will appear in 1978 in the *Journal of Combinatorial Theory*, Series B.

In number theory the most exciting discovery of the past year is that when the transcendental number e is raised to the power of π times $\sqrt{163}$, the result is an integer. The Indian mathematician Srinivasa Ramanujan had conjectured that e to the power of $\pi\sqrt{163}$ is integral in a note in the *Quarterly Journal of Pure*

Unlike most chess-playing programs, MacHic is a learning machine that profits from mistakes, keeping a record of all games in its memory and thus steadily improving. Early in 1974 Pinkleaf started MacHic playing against itself, taking both sides and completing a game on an average of every 1.5 seconds. The machine ran steadily for about seven months.

At the end of the run MacHic announced an extraordinary result. It had established, with a high degree of probability, that pawn to king's rook 4 is a win for White. This was quite unexpected because such an opening move has traditionally been regarded as poor. MacHic could not, of course, make an exhaustive analysis of all possible replies. In constructing a "game tree" for the opening, however, MacHic extended every branch of the tree to a position that any chess master would unhesitatingly judge to be so hopeless for Black that Black should at once resign.

Pinkleaf has been under enormous pressure from world chess leaders to de-

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In May, 1974, John Brillo of the University of Arizona found an ingenious way of applying Euler's constant to the calculation and managed to prove that the number exactly equals 262,537,412,640,768,744. How the prime number 163 manages to convert the expression to an integer is not yet fully understood. Brillo's proof is scheduled to appear in a few years in *Mathematics of Computation*.

An explicit computation

```
GP/PARI CALCULATOR Version 2.8.0 (development 18205-1c269ec)
  i386 running darwin (x86-64 kernel) 64-bit version
compiled: Nov 11 2015, Apple LLVM version 6.0 (clang-600.0.54) (based on LLVM 3.5svn)
      threading engine: single
      (readline not compiled in, extended help enabled)
```

Copyright (C) 2000-2015 The PARI Group

PARI/GP is free software, covered by the GNU General Public License, and comes WITHOUT ANY WARRANTY WHATSOEVER.

Type ? for help, \q to quit.

Type ?15 for how to get moral (and possibly technical) support.

```
parisize = 8000000, primelimit = 500000
? exp(Pi*sqrt(163))
%1 = 262537412640768743.99999999999925007259
? █
```

April 1, 1975 | 1 min read



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Mathematical Games, April 1975

Six sensational discoveries that somehow or another have escaped public attention

BY MARTIN GARDNER

April 1975 Issue 

The Sciences ▾

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[More by Martin Gardner](#)

**SCIENTIFIC
AMERICAN**

This article was published with the title "Mathematical Games" in Scientific American Magazine Vol. 232 No. 4 (April 1975), p. 126

doi:10.1038/scientificamerican0475-126

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The ring $\mathbb{Z}\left[\frac{-1+\sqrt{-163}}{2}\right] = \mathbb{Z}[x]/(x^2 + x + 41)$ is a **principal ideal domain**.

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- ▶ $n^2 + n + 17$ is prime for $0 \leq n \leq 15$;
- ▶ $e^{\pi\sqrt{67}} = 147197952743.999998662454\dots$

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$$1 - 4k \in \{-3, -7, -11, -19, -43, -67, -163\}.$$

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Proved by **Kurt Heegner** in **1952**.

Binary Quadratic Forms: Definition

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These binary quadratic forms are '**equivalent**', and **represent the same integers**.

Binary Quadratic Forms: Discriminant

The **discriminant** of $f(u, v) = au^2 + buv + cv^2$ is

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Simple proof: The latter represents 2 and the former does not.

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- ▶ $h(D)$ is **always finite**.
- ▶ We have $\sum_{0 < -D < X} h(D) \sim cX^{3/2}$.
- ▶ For $D > 0$, the story is similar but more complicated.

Gauss's Composition Law

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Essentially, this was discovered by Gauss.

Dirichlet's Class Number Formula

Let $D < -4$ be a 'fundamental discriminant'.

Theorem (Dirichlet, 1839)

We have

$$h(D) = |D|^{1/2} / \pi \prod_p \left(1 - \left(\frac{D}{p} \right) \frac{1}{p} \right).$$

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Corollary (Siegel, 1935)

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where C is a constant *which we can't possibly determine*.

Prime values of $n^2 + n + 41$

Rabinowicz's Criterion (1913):

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ANDREW GRANVILLE

p is represented by $x^2 + xy + 11y^2$ if and only if $(-43/p) = 1$,
 p is represented by $x^2 + xy + 17y^2$ if and only if $(-67/p) = 1$,
 p is represented by $x^2 + xy + 41y^2$ if and only if $(-163/p) = 1$.

Euler noticed that the polynomial $x^2 + x + 41$ is prime for $x = 0, 1, 2, \dots, 39$, and similarly the other polynomials above. Rabinowicz proved that this is an “if and only if” condition; that is

Rabinowicz's criterion. *We have $h(1 - 4A) = 1$ for $A \geq 2$ if and only if $x^2 + x + A$ is prime for $x = 0, 1, 2, \dots, A - 2$.*

Note that $(A - 1)^2 + (A - 1) + A = A^2$. We will prove Rabinowicz's criterion below

The proof that the above list gives all of the $d < 0$ with $h(d) = 1$ has an interesting history. By 1934 it was known that there is no more than one further such d , but that putative d could not be ruled out by the method. In 1952, Kurt Heegner, a German school teacher proposed an extraordinary proof that there are no further d ; at the time his paper was ignored since it was based on a result from an old book (of Weber) whose proof was known to be incomplete. In 1966 Alan Baker gave a very different proof that was acknowledged to be correct. However, soon afterwards Stark realized that the proofs in Weber are easily corrected, so that Heegner's work had been fundamentally correct. Heegner was subsequently given credit for solving this famous problem, but sadly only after he had died. Heegner's paper contains a most extraordinary construction, widely regarded to be one of the most creative and influential in the history

A. Granville, *Introduction to Gauss's Number Theory*, online notes.

Almost an integer

Let j be the **modular j -function**

$$j(\tau) = q^{-1} + 744 + 196884q + 21493760q^2 + \cdots,$$

defined for $\tau := x + iy$ with $y > 0$, and $q := e^{2\pi i\tau}$.

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It follows by the theory of **complex multiplication** that, for $\tau := \frac{-1+\sqrt{-163}}{2}$, $j(\tau)$ **is an integer**.

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For this τ , we have $q = -e^{-\pi\sqrt{163}}$, so

$$j(\tau) = -e^{\pi\sqrt{163}} + 744 + 196884 \cdot \text{tiny} + 21493760 \cdot \text{tiny}^2 + \cdots \in \mathbb{Z}.$$

The Davenport-Heilbronn Theorem

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Theorem (Davenport-Heilbronn, 1971)

We have

$$N_3(X) = \frac{1}{3\zeta(3)}X + o(X).$$

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Let V be the space of **binary cubic forms**:

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We have

$$\mathrm{Disc}(x) = b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd.$$

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$$\mathrm{Disc}(x) = b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd.$$

Theorem (Levi, Delone-Faddeev, Gan-Gross-Savin)

$G(\mathbb{Z})$ -orbits on $V(\mathbb{Z})$ parametrize **cubic rings**. Further, if $f \leftrightarrow R$,

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$G(\mathbb{Z})$ -orbits on $V(\mathbb{Z})$ parametrize **cubic rings**. Further, if $f \leftrightarrow R$,

- ▶ $\mathrm{Stab}(f)$ is isomorphic to $\mathrm{Aut}(R)$;
- ▶ $\mathrm{Disc}(f) = \mathrm{Disc}(R)$;
- ▶ (Davenport-Heilbronn) R is **maximal** iff, for all primes p , f satisfies a certain congruence condition $(\bmod p^2)$.

Bhargava's Higher Composition Laws

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Theorem

There exists an explicit, "nice" bijection

$$\{ \text{Something nice} \} \longleftrightarrow G(\mathbb{Z}) \backslash V(\mathbb{Z})$$

where V is a f.d. representation of an algebraic group G .

Moreover, **certain arithmetic properties** on the left correspond to **congruence conditions** on the right.

Table 1: Summary of Higher Composition Laws

#	Lattice ($V_{\mathbb{Z}}$)	Group acting ($G_{\mathbb{Z}}$)	Parametrizes ($\mathcal{C}$)	(k)	(n)	(H)
1.	$\{0\}$	-	Linear rings	0	0	A_0
2.	$\tilde{\mathbb{Z}}$	$\mathrm{SL}_1(\mathbb{Z})$	Quadratic rings	1	1	A_1
3.	$(\mathrm{Sym}^2 \mathbb{Z}^2)^*$ (GAUSS'S LAW)	$\mathrm{SL}_2(\mathbb{Z})$	Ideal classes in quadratic rings	2	3	B_2
4.	$\mathrm{Sym}^3 \mathbb{Z}^2$	$\mathrm{SL}_2(\mathbb{Z})$	Order 3 ideal classes in quadratic rings	4	4	G_2
5.	$\mathbb{Z}^2 \otimes \mathrm{Sym}^2 \mathbb{Z}^2$	$\mathrm{SL}_2(\mathbb{Z})^2$	Ideal classes in quadratic rings	4	6	B_3
6.	$\mathbb{Z}^2 \otimes \mathbb{Z}^2 \otimes \mathbb{Z}^2$	$\mathrm{SL}_2(\mathbb{Z})^3$	Pairs of ideal classes in quadratic rings	4	8	D_4
7.	$\mathbb{Z}^2 \otimes \wedge^2 \mathbb{Z}^4$	$\mathrm{SL}_2(\mathbb{Z}) \times \mathrm{SL}_4(\mathbb{Z})$	Ideal classes in quadratic rings	4	12	D_5
8.	$\wedge^3 \mathbb{Z}^6$	$\mathrm{SL}_6(\mathbb{Z})$	Quadratic rings	4	20	E_6
9.	$(\mathrm{Sym}^3 \mathbb{Z}^2)^*$	$\mathrm{GL}_2(\mathbb{Z})$	Cubic rings	4	4	G_2
10.	$\mathbb{Z}^2 \otimes \mathrm{Sym}^2 \mathbb{Z}^3$	$\mathrm{GL}_2(\mathbb{Z}) \times \mathrm{SL}_3(\mathbb{Z})$	Order 2 ideal classes in cubic rings	12	12	F_4
11.	$\mathbb{Z}^2 \otimes \mathbb{Z}^3 \otimes \mathbb{Z}^3$	$\mathrm{GL}_2(\mathbb{Z}) \times \mathrm{SL}_3(\mathbb{Z})^2$	Ideal classes in cubic rings	12	18	E_6
12.	$\mathbb{Z}^2 \otimes \wedge^2 \mathbb{Z}^6$	$\mathrm{GL}_2(\mathbb{Z}) \times \mathrm{SL}_6(\mathbb{Z})$	Cubic rings	12	30	E_7
13.	$(\mathbb{Z}^2 \otimes \mathrm{Sym}^2 \mathbb{Z}^3)^*$	$\mathrm{GL}_2(\mathbb{Z}) \times \mathrm{SL}_3(\mathbb{Z})$	Quartic rings	12	12	F_4
14.	$\mathbb{Z}^4 \otimes \wedge^2 \mathbb{Z}^5$	$\mathrm{GL}_4(\mathbb{Z}) \times \mathrm{SL}_5(\mathbb{Z})$	Quintic rings	40	40	E_8

Bhargava, *Higher composition laws IV*, Ann. Math., 2008

Prehomogeneous Vector Spaces

Over $\mathbb{C}$, the preceding are all **prehomogeneous vector spaces**:

M. Sato and T. Kimura
Nagoya Math. J.
Vol. 65 (1977) 1–155

A CLASSIFICATION OF IRREDUCIBLE PREHOMOGENEOUS VECTOR SPACES AND THEIR RELATIVE INVARIANTS

M. SATO AND T. KIMURA*

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We call such a triplet (G, ρ, V) a *prehomogeneous vector space* if V has a Zariski-dense G -orbit. The main purpose of this paper is to classify all prehomogeneous vector spaces when ρ is irreducible, and to investigate their relative invariants and the regularity.

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Also a **zeta function theory** due to **Sato** and **Shintani**.

Bhargava's Averaging Method

Suppose that **blah, blah, blah...**

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Let $N(X)$ count the $G(\mathbb{Z})$ -orbits on $V(\mathbb{Z})$ with $0 < |\text{Disc}(x)| < X$.

Lemma (Bhargava)

We have

$$\begin{aligned} N(X) &= \frac{\int_{v \in B} \#\{x \in \mathcal{F}_G v \cap V(\mathbb{Z}) : |\text{Disc}(x)| < X\} d^*v}{n \int_{v \in B} d^*v} \\ &= C \int_{g \in \mathcal{F}_G} \#\{x \in gB \cap V(\mathbb{Z}) : |\text{Disc}(x)| < X\} dg, \end{aligned}$$

where **PLEASE DONT ACTUALLY READ THIS:**

- ▶ $\mathcal{F}_G$ is a fundamental domain for the action of $G(\mathbb{Z})$ on $G(\mathbb{R})$;
- ▶ $B \subseteq V(\mathbb{R})$ is a compact, connected region supported away from the singular set $\{v \in V(\mathbb{R}) : \text{Disc}(v) = 0\}$;
- ▶ d^*v is a measure on $V(\mathbb{R})$ invariant by the action of $G(\mathbb{R})$;
- ▶ The multiset $G(\mathbb{R})v$ is an n -fold cover of $V(\mathbb{R})$ for $v \in B$.

Some Recent Trends in Arithmetic Statistics

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1. Even More Parametrizations

	Group (s.s.)	Representation	Geometric Data	Invariants	Dynkin	§
1.	SL_2	$Sym^4(2)$	(C, L_2)	2, 3	$A_2^{(2)}$	4.1
2.	SL_2^2	$Sym^2(2) \otimes Sym^2(2)$	$(C, L_2, L'_2) \sim (C, L_2, P)$	2, 3, 4	$D_3^{(2)}$	6.1
3.	SL_2^4	$2 \otimes 2 \otimes 2 \otimes 2$	$(C, L_2, L'_2, L''_2) \sim (C, L_2, P, P')$	2, 4, 4, 6	$D_4^{(1)}$	6.2
4.	SL_2^3	$2 \otimes 2 \otimes Sym^2(2)$	$(C, L_2, L'_2) \sim (C, L_2, P)$	2, 4, 6	$B_3^{(1)}$	6.3.1
5.	SL_2^2	$Sym^2(2) \otimes Sym^2(2)$	$(C, L_2, L'_2) \sim (C, L_2, P)$	2, 3, 4	$D_3^{(2)}$	6.3.3
6.	SL_2^2	$2 \otimes Sym^3(2)$	(C, L_2, P_3)	2, 6	$G_2^{(1)}$	6.3.2
7.	SL_2	$Sym^4(2)$	(C, L_2, P_3)	2, 3	$A_2^{(2)}$	6.3.4
8.	$SL_2^2 \times GL_4$	$2 \otimes 2 \otimes \wedge^2(4)$	$(C, L_2, M_{2,6})$	2, 4, 6, 8	$D_5^{(1)}$	6.6.1
9.	$SL_2 \times SL_6$	$2 \otimes \wedge^3(6)$	$(C, L_2, M_{3,6})$ with $L^{\otimes 3} \cong \det M$	2, 6, 8, 12	$E_6^{(1)}$	6.6.2
10.	$SL_2 \times Sp_6$	$2 \otimes \wedge_0^3(6)$	$(C, L_2, (M_{3,6}, \varphi))$ with $L^{\otimes 3} \cong \det M$	2, 6, 8, 12	$E_6^{(2)}$	6.6.3
11.	$SL_2 \times Spin_{12}$	$2 \otimes S^+(32)$	$(C \rightarrow \mathbb{P}^1(\mathcal{H}_3(\mathbb{H})), L_2)$	2, 6, 8, 12	$E_7^{(1)}$	6.6.3
12.	$SL_2 \times E_7$	$2 \otimes 56$	$(C \rightarrow \mathbb{P}^1(\mathcal{H}_3(\mathbb{O})), L_2)$	2, 6, 8, 12	$E_8^{(1)}$	6.6.3
13.	SL_3	$Sym^3(3)$	(C, L_3)	4, 6	$D_4^{(3)}$	4.2
14.	SL_3^3	$3 \otimes 3 \otimes 3$	$(C, L_3, L'_3) \sim (C, L_3, P)$	6, 9, 12	$E_6^{(1)}$	5.1
15.	SL_3^3	$3 \otimes Sym^2(3)$	(C, L_3, P_2)	6, 12	$F_4^{(1)}$	5.2.1
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17.	$SL_3 \times SL_6$	$3 \otimes \wedge^2(6)$	$(C, L_3, M_{2,6})$ with $L^{\otimes 2} \cong \det M$	6, 12, 18	$E_7^{(1)}$	5.5
18.	$SL_3 \times E_6$	$3 \otimes 27$	$(C \hookrightarrow \mathbb{P}^2(\mathbb{O}), L_3)$	6, 12, 18	$E_8^{(1)}$	5.4
19.	$SL_2 \times SL_4$	$2 \otimes Sym^2(4)$	(C, L_4)	8, 12	$E_6^{(2)}$	4.3
20.	$SL_5 \times SL_5$	$\wedge^2(5) \otimes 5$	(C, L_5)	20, 30	$E_8^{(1)}$	4.4

Table 1: Table of coregular representations and their moduli interpretations

Bhargava and Ho, *Coregular spaces and genus one curves*, Camb. J. Math

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Theorem (Bhargava-Shankar 2010-2013)

The average rank of elliptic curves is < 0.885 .

2. Getting Better at Counting

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- ▶ Multiple Dirichlet Series
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4. Making Predictions With Stacks

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Conjecture (Malle 2002, 2004; Loughran-Santens 2024)

We have

$$N_d(X, G) = c(G)X^{1/a(G)}(\log X)^{b(G)-1},$$

where $a(G)$ and $b(G)$ are **positive integers**, and $c(G)$ is **positive real**.

5. Topology and Function Fields

Do arithmetic statistics over $\mathbb{F}_q[t]$ instead of $\mathbb{Z}$.

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Far reaching consequences: Malle's conjecture, Cohen-Lenstra-Martinet heuristics, ...

6. What will the near future bring?

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Watch this space!